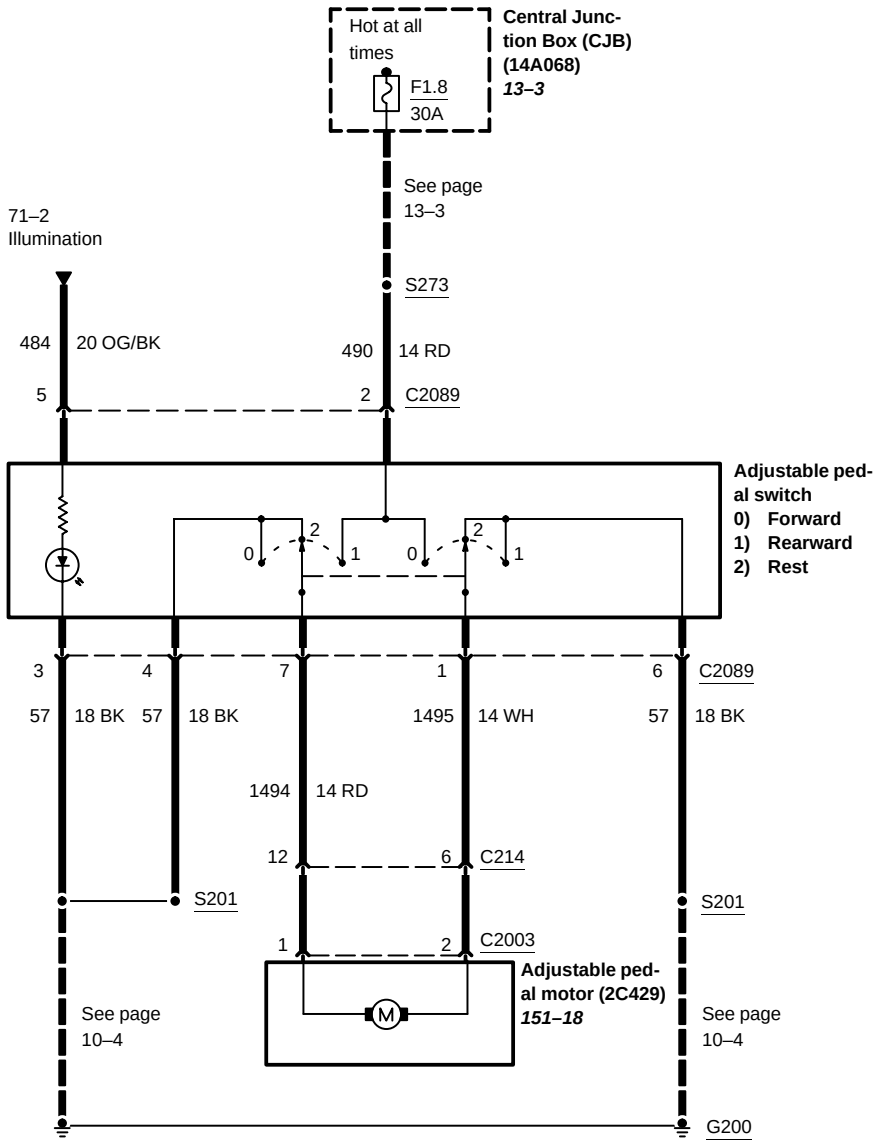without memory